

Self-Reflection 2

Every time you change the x value, it chooses one of the predictions depending on the number you write. The bigger the number, the lowest the probability that it would fit, meaning that it may make less sense. $X = 1$ chooses the most likely valid choice but the further away you choose, doesn't mean that it will be random but that it has the lowest probability of making sense within the context. I struggled to make it make sense with what

Every time I change the x value, the program chooses one of the predictions depending on the number you write. When $x = 1$, it chooses the most probable option, so the result makes more sense within the context of the line. As the x value increases, the program goes further down the list of predictions. This means that the selected token has a lower probability based on the model and can make the poem sound stranger or less connected to the original sentence. However, a larger x value does not mean that the result is random. GPT-2 is still ranking its predictions based on the context, I am just choosing an option that the model considers less likely.

One thing I struggled with was connecting this to the first part of the assignment, where I found the 10 most probable next tokens. My P+7 code was giving me different results from my initial view and expectations based on the probability exercise. I realized that the way I filtered the predictions affected the result. In my P+x code, I used `isalpha()` so that symbols and punctuation would not be counted as possible replacements. This means that P+7 in my code represents the seventh valid alphabetic prediction, rather than necessarily the seventh token in GPT-2's original ranking.

To implement a P+7 technique where all nouns are replaced, I would have to identify which words in the poem are nouns. I could use a part-of-speech tagging library to detect them. Then, for each noun I would give GPT-2 the text that comes before that noun and get the predicted next tokens. I would filter the predictions to keep alternative words that make sense and select the seventh-highest valid prediction. Then, I would replace the original noun while leaving the other words unchanged. This would apply the same P+7 idea with the specifications throughout the poem.